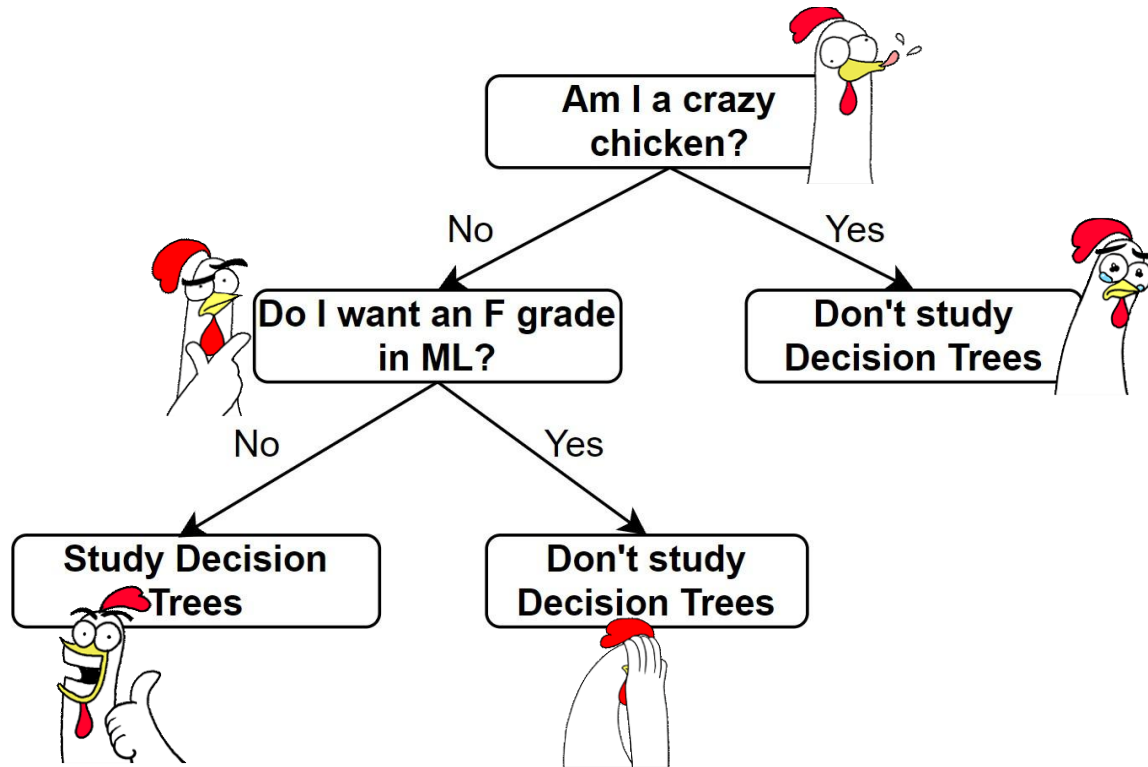


# **Machine Learning**

**Instructor: Hafiz Abdul Rehman**

**Lecture 22: Decision Tree**

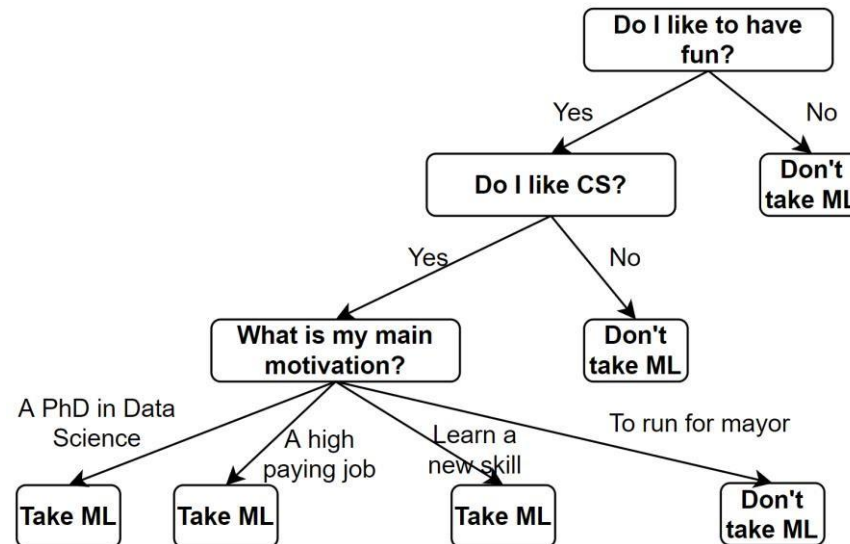
# Why do chickens learn decision trees?



Because it was not crazy **AND** it did not want an F grade in ML!



# To be or not to be a (Machine Learner) Bee!



Take ML if you “like to have fun” **AND** “like CS” **AND** (want a PhD in DS **OR** a high paying job **OR** to learn a new skill) =

$$\text{Like Fun} \wedge \text{Like CS} \wedge (\text{PhD} \vee \text{Pay} \vee \text{Skill}) = \\ (\text{Like Fun} \wedge \text{Like CS} \wedge \text{PhD}) \vee (\text{Like Fun} \wedge \text{Like CS} \wedge \text{Pay}) \vee (\text{Like Fun} \wedge \text{Like CS} \wedge \text{Skill})$$

- Highly intuitive and explainable model
- Read the concise description of what makes an item positive off the tree
  - No black boxes
- Disjunctive Normal Form (DNF) – is a universal approximator!
  - The NOT operation can simply be added by asking a negative question e.g. “having a skill” vs “not having a skill”



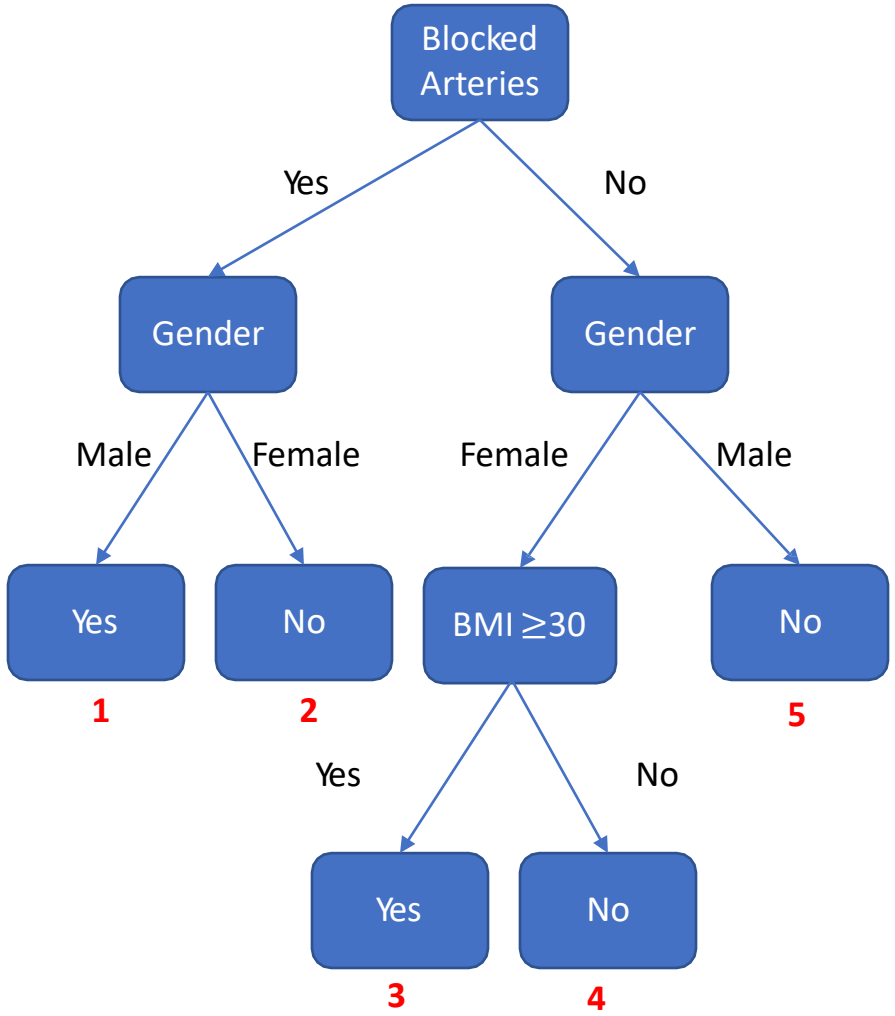
# Decision Trees for Classification and Regression

| Age | Blocked Arteries | Height (inches) | Gender | BMI | Heart Disease | Threat (%) |
|-----|------------------|-----------------|--------|-----|---------------|------------|
| 23  | No               | 67              | M      | 35  | No            | 47         |
| 45  | Yes              | 61              | F      | 34  | No            | 45         |
| 37  | Yes              | 68              | M      | 23  | Yes           | 60         |
| 56  | Yes              | 65              | M      | 36  | Yes           | 78         |
| 75  | No               | 67              | F      | 37  | Yes           | 85         |
| 63  | No               | 69              | F      | 39  | Yes           | 90         |
| 72  | Yes              | 64              | F      | 22  | No            | 20         |
| 21  | No               | 62              | F      | 23  | No            | 23         |
| 67  | Yes              | 72              | M      | 19  | Yes           | 55         |
| 49  | No               | 70              | M      | 40  | No            | 49         |



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|-----|------------------|-----------------|--------|-----|---------------|------------|
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| 21  | No               | 62              | F      | 23  | No            | 23         |
| 67  | Yes              | 72              | M      | 19  | Yes           | 55         |
| 49  | No               | 70              | M      | 40  | No            | 49         |

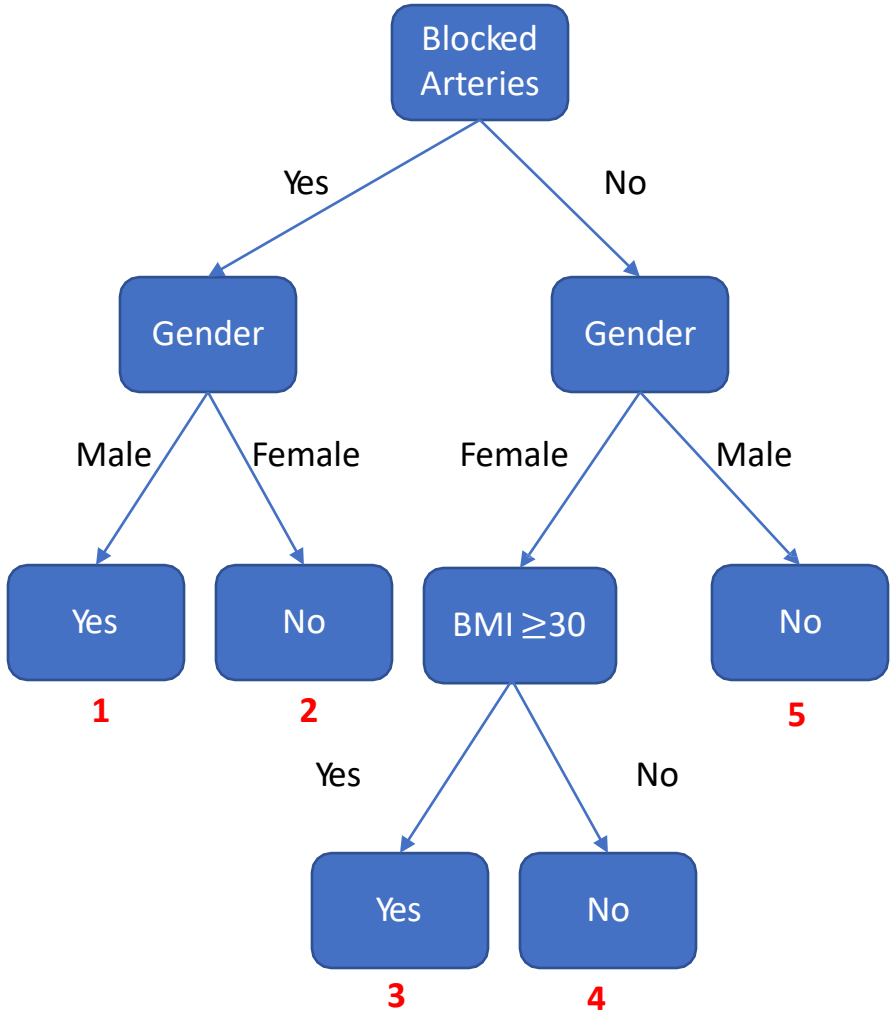


| Age | Blocked Arteries | Height (inches) | Gender | BMI | Heart Disease | Threat (%) |
|-----|------------------|-----------------|--------|-----|---------------|------------|
| 25  | No               | 63              | M      | 23  |               |            |
| 34  | Yes              | 70              | M      | 33  |               |            |
| 50  | No               | 65              | F      | 33  |               |            |



# Decision Trees for Classification and Regression

| Age | Blocked Arteries | Height (inches) | Gender | BMI | Heart Disease | Threat (%) |   |
|-----|------------------|-----------------|--------|-----|---------------|------------|---|
| 23  | No               | 67              | M      | 35  | No            | 47         | 5 |
| 45  | Yes              | 61              | F      | 34  | No            | 45         | 2 |
| 37  | Yes              | 68              | M      | 23  | Yes           | 60         | 1 |
| 56  | Yes              | 65              | M      | 36  | Yes           | 78         | 1 |
| 75  | No               | 67              | F      | 37  | Yes           | 85         | 3 |
| 63  | No               | 69              | F      | 39  | Yes           | 90         | 3 |
| 72  | Yes              | 64              | F      | 22  | No            | 20         | 2 |
| 21  | No               | 62              | F      | 23  | No            | 23         | 4 |
| 67  | Yes              | 72              | M      | 19  | Yes           | 55         | 1 |
| 49  | No               | 70              | M      | 40  | No            | 49         | 5 |

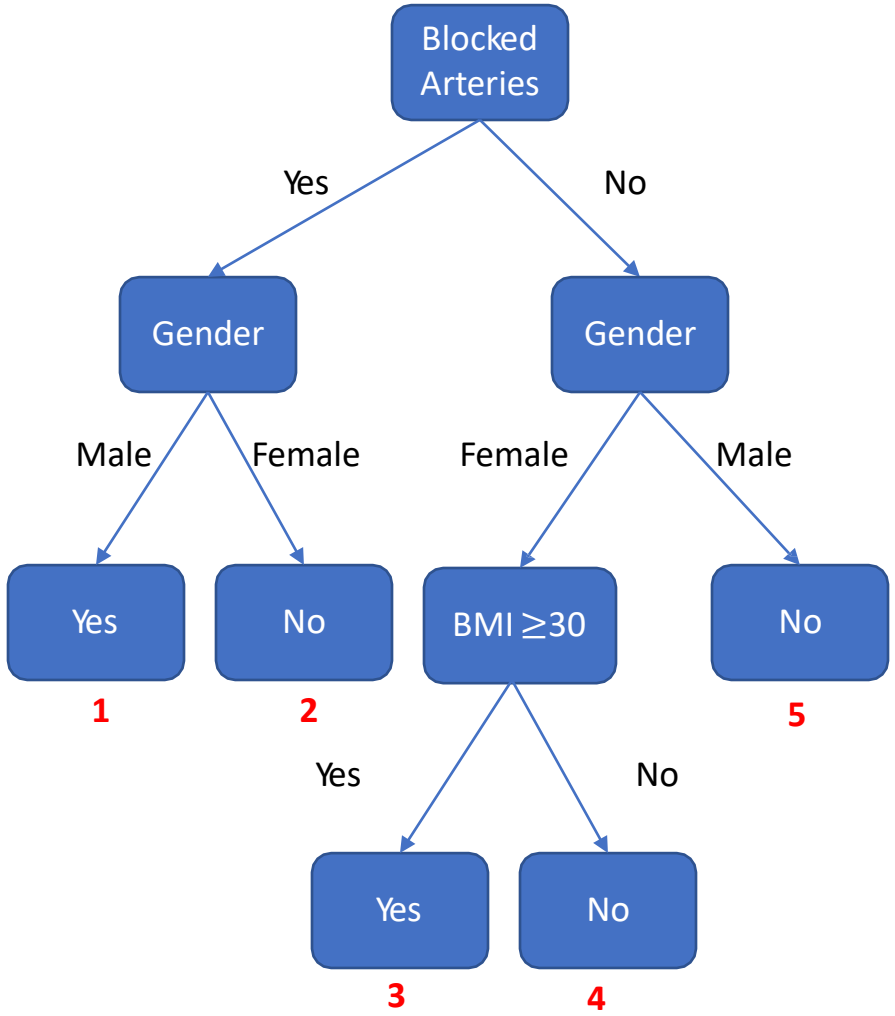


| Age | Blocked Arteries | Height (inches) | Gender | BMI | Heart Disease | Threat (%) |
|-----|------------------|-----------------|--------|-----|---------------|------------|
| 25  | No               | 63              | M      | 23  |               |            |
| 34  | Yes              | 70              | M      | 33  |               |            |
| 50  | No               | 65              | F      | 33  |               |            |



# Decision Trees for Classification and Regression

| Age | Blocked Arteries | Height (inches) | Gender | BMI | Heart Disease | Threat (%) |   |
|-----|------------------|-----------------|--------|-----|---------------|------------|---|
| 23  | No               | 67              | M      | 35  | No            | 47         | 5 |
| 45  | Yes              | 61              | F      | 34  | No            | 45         | 2 |
| 37  | Yes              | 68              | M      | 23  | Yes           | 60         | 1 |
| 56  | Yes              | 65              | M      | 36  | Yes           | 78         | 1 |
| 75  | No               | 67              | F      | 37  | Yes           | 85         | 3 |
| 63  | No               | 69              | F      | 39  | Yes           | 90         | 3 |
| 72  | Yes              | 64              | F      | 22  | No            | 20         | 2 |
| 21  | No               | 62              | F      | 23  | No            | 23         | 4 |
| 67  | Yes              | 72              | M      | 19  | Yes           | 55         | 1 |
| 49  | No               | 70              | M      | 40  | No            | 49         | 5 |



| Age | Blocked Arteries | Height (inches) | Gender | BMI | Heart Disease | Threat (%) |
|-----|------------------|-----------------|--------|-----|---------------|------------|
| 25  | No               | 63              | M      | 23  | NO            | 48         |
| 34  | Yes              | 70              | M      | 33  | Yes           | 64.3       |
| 50  | No               | 65              | F      | 33  | Yes           | 87.5       |

Might be incorrect as the BMI is low. So we need to grow a separate tree for regression.

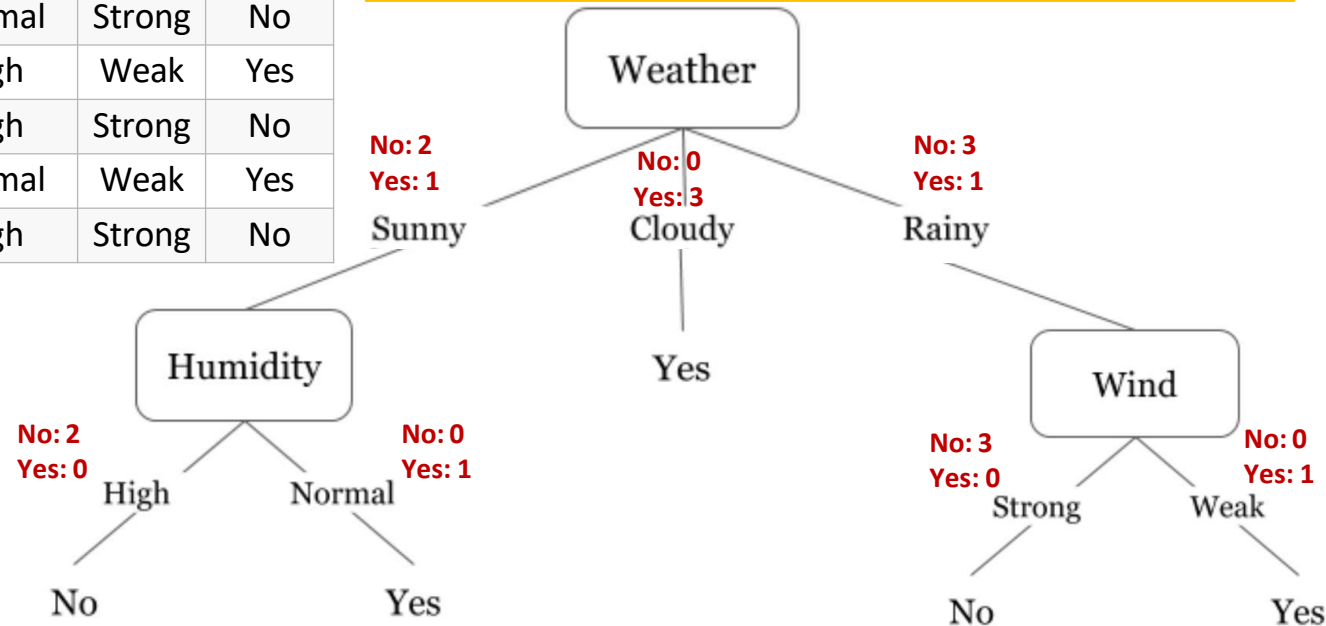
Voting in case of impure leaves.



# Time for a real example – To play golf or not

| Day | Weather | Temperature | Humidity | Wind   | Play? |
|-----|---------|-------------|----------|--------|-------|
| 1   | Sunny   | Hot         | High     | Weak   | No    |
| 2   | Cloudy  | Hot         | High     | Weak   | Yes   |
| 3   | Sunny   | Mild        | Normal   | Strong | Yes   |
| 4   | Cloudy  | Mild        | High     | Strong | Yes   |
| 5   | Rainy   | Mild        | High     | Strong | No    |
| 6   | Rainy   | Cool        | Normal   | Strong | No    |
| 7   | Rainy   | Mild        | High     | Weak   | Yes   |
| 8   | Sunny   | Hot         | High     | Strong | No    |
| 9   | Cloudy  | Hot         | Normal   | Weak   | Yes   |
| 10  | Rainy   | Mild        | High     | Strong | No    |

- If impure leaves are allowed, the test label can be decided using voting
- The features could also be real-valued
- The tree does not need to be binary
- We prefer the shallowest (simplest tree) that gets the job done!

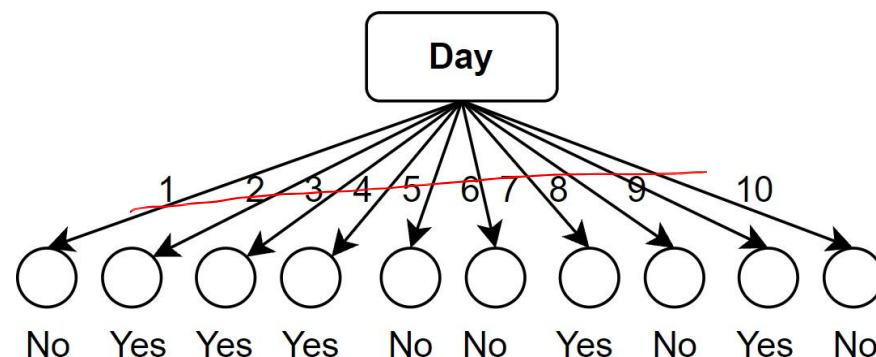


|  | Weather | Temperature | Humidity | Wind   | Play? |
|--|---------|-------------|----------|--------|-------|
|  | Cloudy  | Mild        | High     | Strong | ?     |
|  | Rainy   | Mild        | Normal   | Strong | ?     |



# How about this tree? Short, sweet and perfect?

| Day | Weather | Temperature | Humidity | Wind   | Play? |
|-----|---------|-------------|----------|--------|-------|
| 1   | Sunny   | Hot         | High     | Weak   | No    |
| 2   | Cloudy  | Hot         | High     | Weak   | Yes   |
| 3   | Sunny   | Mild        | Normal   | Strong | Yes   |
| 4   | Cloudy  | Mild        | High     | Strong | Yes   |
| 5   | Rainy   | Mild        | High     | Strong | No    |
| 6   | Rainy   | Cool        | Normal   | Strong | No    |
| 7   | Rainy   | Mild        | High     | Weak   | Yes   |
| 8   | Sunny   | Hot         | High     | Strong | No    |
| 9   | Cloudy  | Hot         | Normal   | Weak   | Yes   |
| 10  | Rainy   | Mild        | High     | Strong | No    |



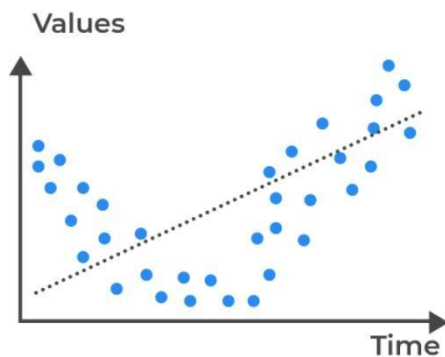
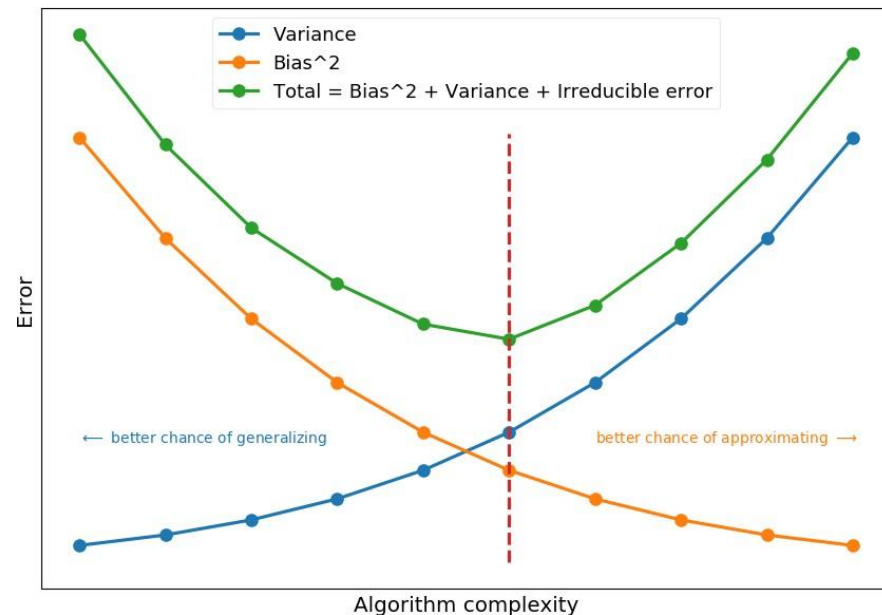
- This is a very specific form of overfitting!
- This tree is a perfect classifier for the training data but will never work for any new data
- We would need ways to prevent such trees from happening
- We do so by discouraging leaves with very small number of instances

|    | Weather | Temperature | Humidity | Wind   | Play? |
|----|---------|-------------|----------|--------|-------|
| 13 | Cloudy  | Mild        | High     | Strong | ?     |
| 19 | Rainy   | Mild        | Normal   | Strong | ?     |

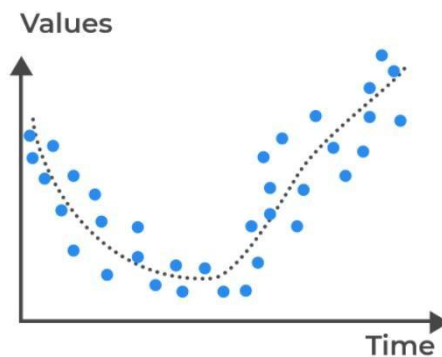


# Bias and Variance

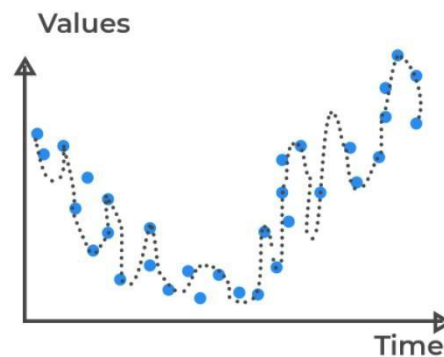
- **Bias** is the error that comes from wrong model assumptions, which do not allow an algorithm to learn all patterns from training data.
- **Variance** is the error that comes from model sensitivity to features specific to the training data (e.g., noise).
- **High bias** usually leads to **underfitting** and **high variance** usually leads to **overfitting**.
- Total error also depends on irreducible error (noise that can not be reduced by algorithm)
- Our Ultimate goal is to minimize the total error



**Underfitted**  
(High bias error)



**Good Fit/Robust**  
(Balance between bias and variance)

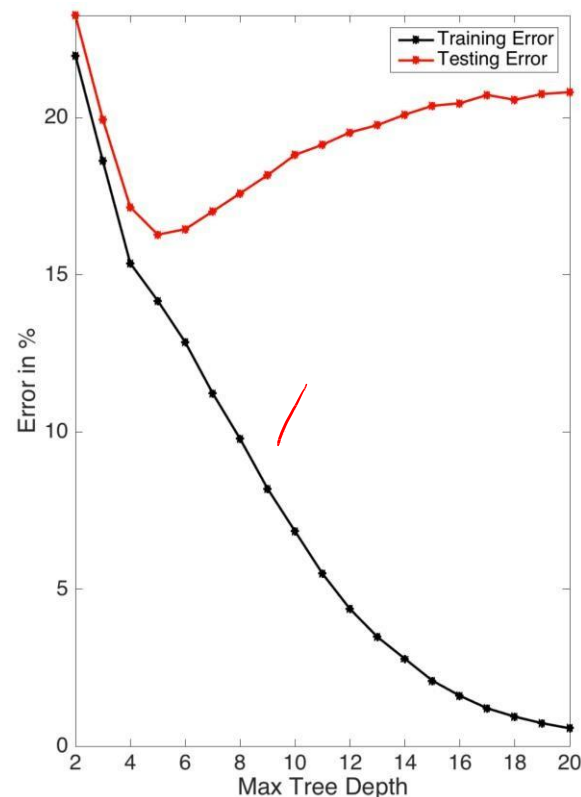


**Overfitted**  
(High variance error)



# Bias and Variance – Height of Tree

- Decision trees are sensitive to splits – small changes in training data may change a tree structure
- Shallow trees have high bias – underfitting – and low variance
- Deeper trees have high variance – overfitting – and low bias
  - We can always add a levels to perfectly fit any training data and even model the noise
    - Unless the dataset is inconsistent – instances having different labels for the same inputs
    - This leads to very high variance
    - The leaves may just have singletons left in them
- **Solutions (to be discussed later):**
  - Reduce **variance** using **Bagging** (Random Forests)
  - Reduce **bias** using **Boosting** (ada boost, gradient boosted trees)

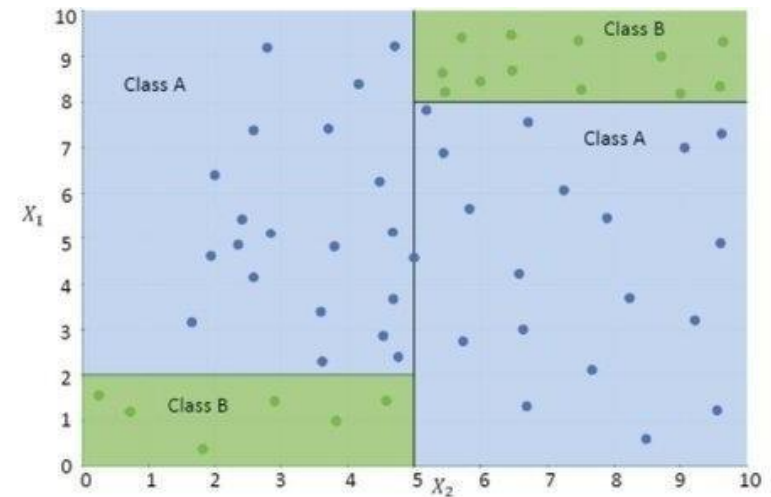


# Partitioning of feature space – Real numbered attributes

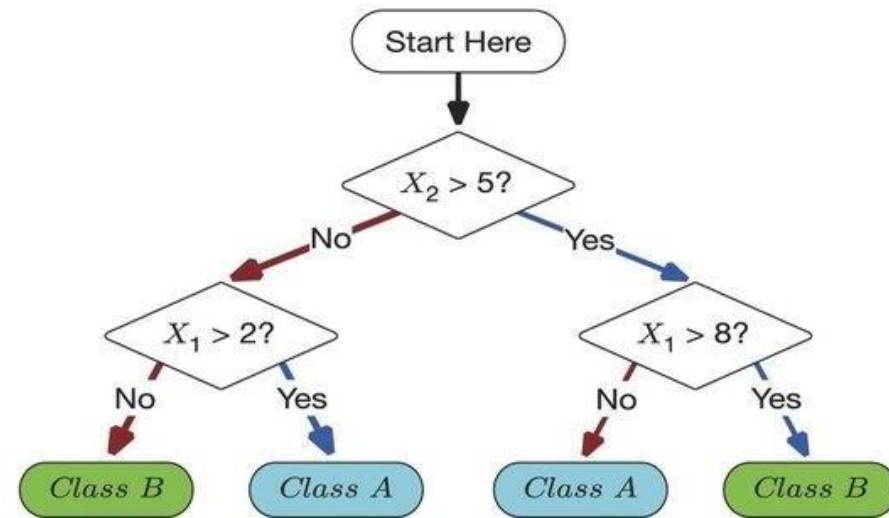
- **if**  $x_2 > 5$ :
  - **if**  $x_1 < 2$ :
    - Class A
  - **else**:
    - Class B
- **else**:
  - **if**  $x_1 > 8$ :
    - Class B
  - **else**:
    - Class A

- For each split we create axis-aligned hyperplanes in n-Dimensions
- For **real-valued attributes**, we need to define **thresholds**

a)

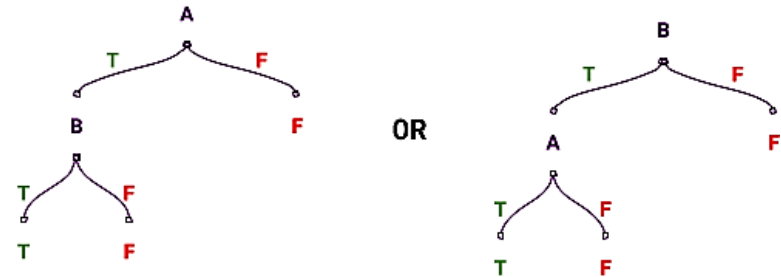


b)

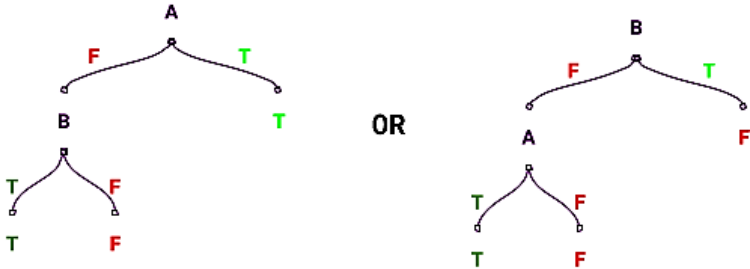


# Boolean functions

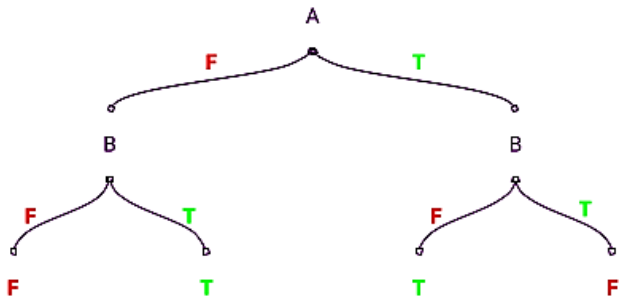
| A | B | A AND B |
|---|---|---------|
| F | F | F       |
| F | T | F       |
| T | F | F       |
| T | T | T       |



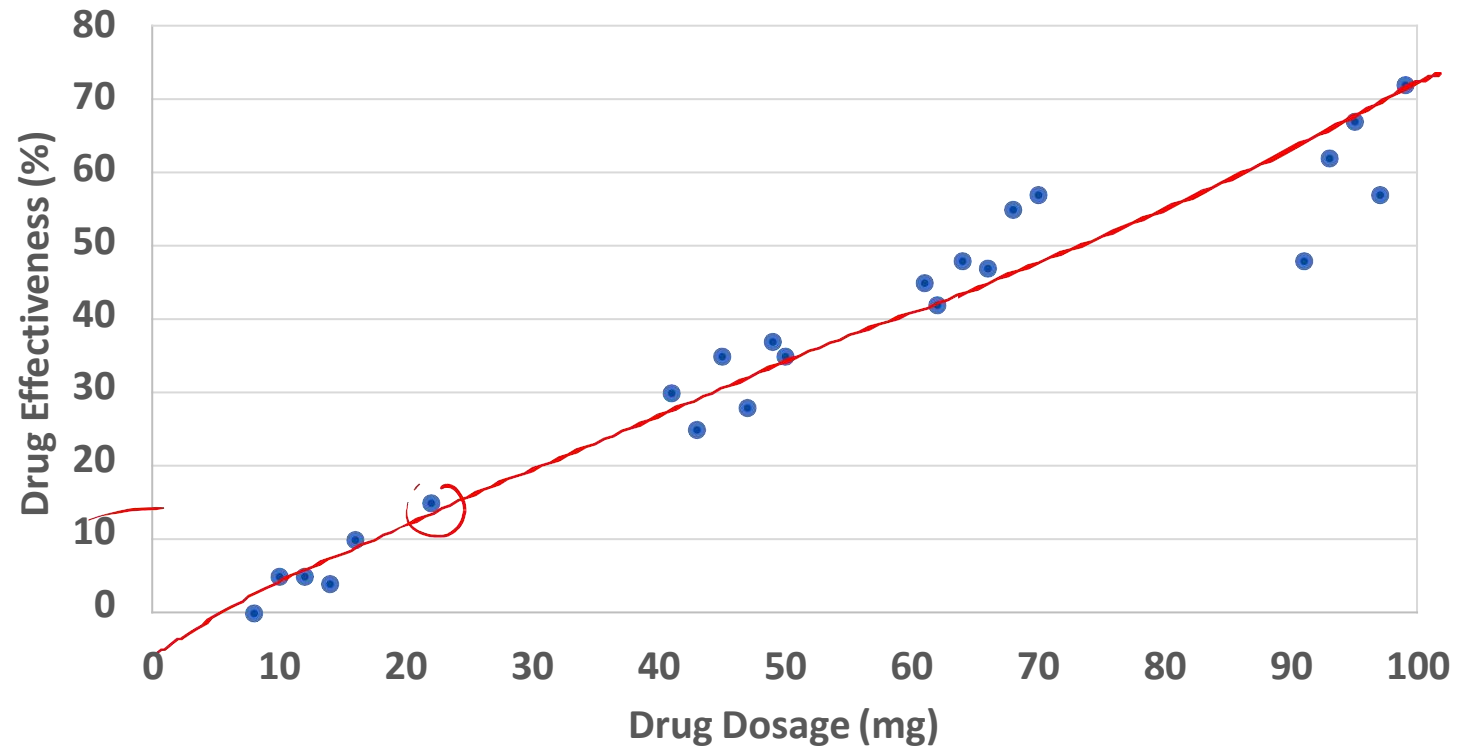
| A | B | A OR B |
|---|---|--------|
| F | F | F      |
| F | T | T      |
| T | F | T      |
| T | T | T      |



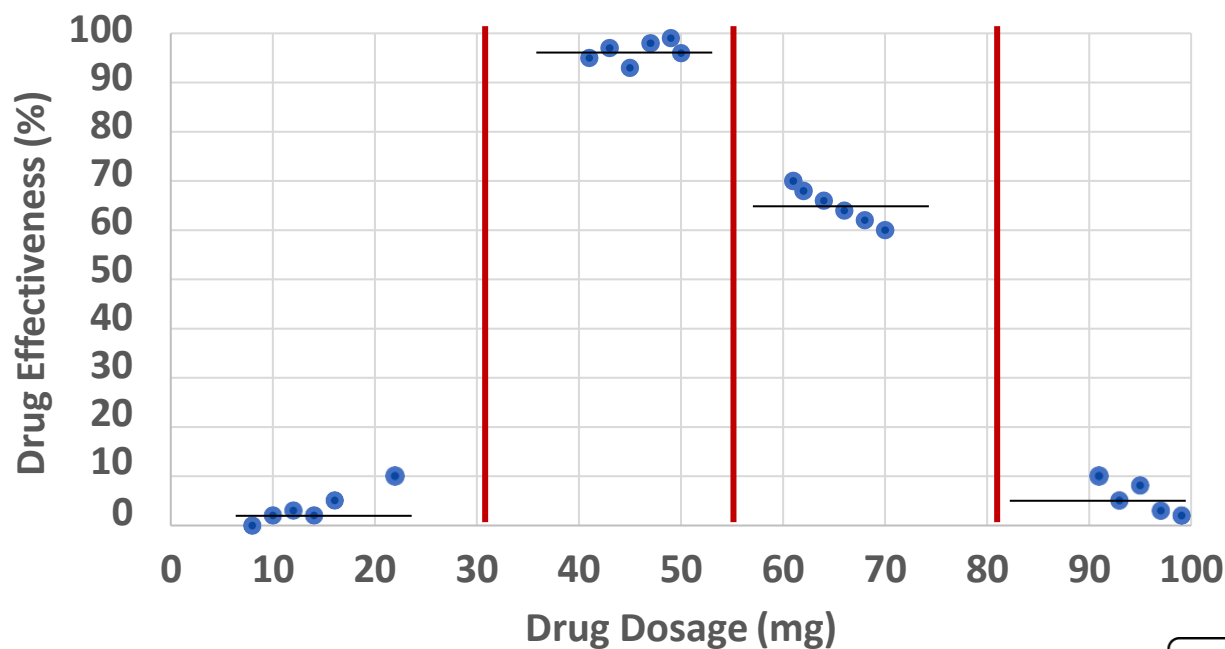
| A | B | A XOR B |
|---|---|---------|
| F | F | F       |
| F | T | T       |
| T | F | T       |
| T | T | F       |



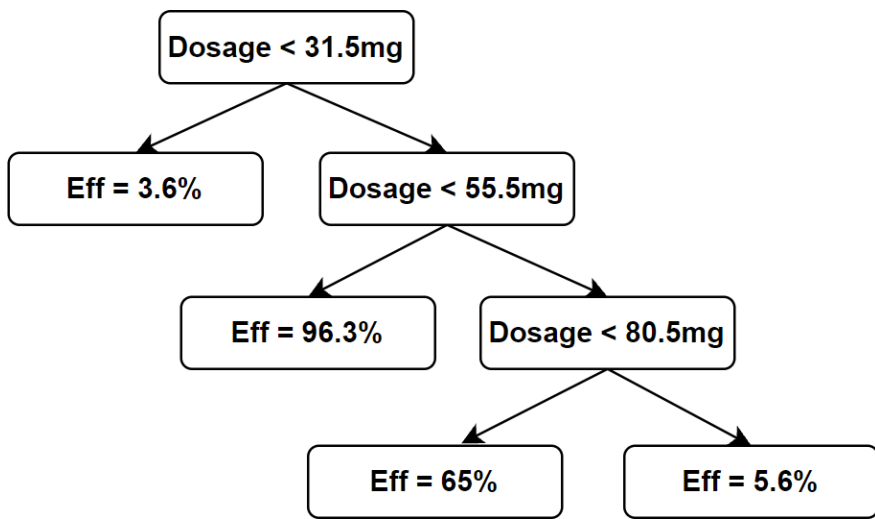
# Regression using decision trees



# Regression using decision trees



New datapoint: dosage = 70mg



**Thank you!**

